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Discussion Agenda 12/13/2013

Below is a discussion agenda / live progress report in an approximate order of importance for the 4th Web Conference of the AGORA Project (Dec. 13, 2013, 9am PST/6pm CET).

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[1. Progress Report \(as of Dec. 2013\)](#)

We are actively making a lot of progress, chiefly, in three different fronts below (numbered tentatively). If you wish to contribute to any of these efforts, please talk to the respective leaders.

We highly encourage *open* and *public* communication within the Collaboration. Each Paper group or Working Group may make use of the Project mailing list, SeeVogh, Google Groups/Hangouts, or mini-workshops (see the opportunity at bottom of this page) to coordinate efforts within. Each group is also asked to report their progress in their respective Workspace so any Collaboration member can follow what is going on.

	PAPER "2"	PAPER "3"	PAPER "4"																																																
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo)	• Implement and test common astrophysics package • Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Calibrate SF/Fbck recipes using common disk ICs																																																
LEADERS	• Oliver Hahn (ETH)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (Edinburgh)	• Oscar Agertz (Surrey) • Romain Teyssier (Zurich) • James Wadsley (McMaster)																																																
PARTICIPANTS	• Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - VI (dwarf galaxies) • People in charge:	• Working Groups engaged: - I (common physics) - III (common cosmo ICs) • People in charge:	• Working Groups engaged: - II (common disk ICs) - IV (common analysis) - V (subgrid physics) • People in charge:																																																
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DELIVERABLES	• Push out PAPER "2"	• Push out PAPER "3"	• Push out PAPER "4"																																																

IN 2014			
PROGRESS MADE OR BEING MADE IN 2013	• ~300 pc res. run completed for the Flagship paper. • ~100 pc res. run to be performed with newly defined cosmological IC of a $\sim 1e11$ Ms halo (MUSIC bug fixed; new IC .conf files to be distributed via Paper 2's Workspace and emails). • yt – Enzo interface for ellipsoidal ICs to be sorted out. • yt-3.0 – Rockstar interface is being improved (discussion on how to use it as of Nov. 2013: here).	• 3-step approach to initiate hydrodynamics runs: [Step 1] DM-only run [Step 2] Hydrodynamic run at $\sim 0.5 - 1$ kpc res. + Vanilla physics (UV bkgrnd, GRACKLE cooling, adiabatic EoS above n_{crit}, <i>but no subgrid physics</i>, i.e. no SF or Fbck) [Step 3] Production run at ~ 100 pc res. + common physics + common parameters • Common physics including GRACKLE being implemented in ART-II, ENZO, GADGET3 and GASOLINE. • Britton Smith is working to add a full equilibrium cooling solver to GRACKLE, in addition to its existing non-equilibrium solver. Ideally, it will not require much additional coding for those groups that have already done the work to implement GRACKLE in their codes.	• Isolated disk ICs read and run using ART-II, ENZO, GADGET3, GASOLINE and RAMSES; pilot studies going well so far. • Need to sort out a equilibrium cooling table in GRACKLE. • Outputs being analyzed with common yt scripts. This group is driving the development effort to visualize gas in yt across all participating codes. • yt-3.0 works nearly completely for SPH codes. The GADGET code is being tested by WG IV and the community contacts for GADGET/GASOLINE codes, for a correct smoothing length for grid cells. • Future roadmap for yt-3.0 includes (in an approximate order of priority): units refactoring, rebranding (including many API changes), scaling n-body datasets, and documentation.
TIMELINE IN 2014	• (to be filled)	• (to be filled)	• (to be filled)
SUBSCRIBE TO UP-TO-DATE PROGRESS	• Please see the progress report in Paper 2's Workspace and/or in WG III's workspace. • MUSIC Google group	• Please see the progress report in Paper 3's Workspace • Please see this page for more information on implementing the GRACKLE cooling, and join the GRACKLE Google group.	• Please see the progress report in Paper 4's Workspace and/or in WG V's workspace. • Please also join the yt-AGORA Google group for extensive discussions on applying yt for cross-code comparison. • yt Users/Dev Community

2. Clarifying the AGORA Common Physics

A few comments and clarifications about the GRACKLE equilibrium and non-equilibrium cooling/chemistry for the AGORA Common Physics Package (extensively discussed and carefully worded by the Steering Committee, Dec. 2013).

- (1) No substantial modifications appears to be needed to the text of the Flagship paper. Here we state that "... GRACKLE provides a non-equilibrium primordial chemistry network for atomic H and He ...", but it does not explicitly say that the AGORA simulations will always be run in a non-equilibrium cooling mode. We would suggest not be too specific here (something we may regret later) and simply change it to "... GRACKLE provides a full non-equilibrium primordial chemistry network for atomic H and He as well as a look-up table for equilibrium cooling ... Depending on the problem at hand, both solvers may be used by the AGORA simulations" and leave it at that.^[1]
- (2) We should definitely track gas metallicity in all the AGORA cosmological simulations (as stated in the Flagship paper!). There are large differences in the outcome of a Eris-type simulation run with primordial vs. metal cooling. And we do want the AGORA simulations to be state-of-the-art. As stated in the Flagship paper about GRACKLE "... All metal cooling rates are tabulated for solar abundances as a function of total hydrogen number density, gas temperature, and redshift ... and are scaled linearly with metallicity ..." So there is no need to do any interpolation!
- (3) In the Proof-of-Concept-NOSUBGRID run (**Paper "3"**) and in the isolated disk comparison (**Paper "4"**), we could use the equilibrium cooling table as the default option. A few of the codes could be run in non-equilibrium as well as equilibrium cooling mode so we could compare the results and check whether non-equilibrium cooling makes a significant difference and if so where and when. These should *not* be computationally expensive calculations. For this, we will deeply appreciate Britton Smith and Working Group I for their effort to provide us with the aforementioned equilibrium cooling table. Depending on which paper(s) each code group participates, it may be asked to implement the non-equilibrium GRACKLE interface and/or an equilibrium look-up routine.
- (4) For the cosmological big runs, (**Papers "5", "6", etc.**) we suggest that we use whatever makes sense following the results of Papers "3" and "4". We can discuss this further later.

These should remove any confusion about the cooling and chemistry in the AGORA Projects. With the full consent in the meeting today, **it will be assumed from now on that the entire group is in full agreement with the plan that has just been laid out.** This is critical as we do *not* want to let participants waste their manpower and resources in implementing things that won't be utilized in the Project.

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[1] Piero and Ji-hoon will accordingly rework the text of Section 3.1 of the Flagship paper so that our intended path is clearly described. (Coincidentally, the Flagship paper has now been accepted by the ApJS as of Nov. 2013; for more info, see below. We are sure we will have a chance to edit the text in the editorial process ahead.)

[3. Live AGORA Simulations Tracker](#)

- When we finally start running production runs using cosmological ICs, let us keep track of the available simulation outputs in the Collaboration so that science-oriented WGs can better coordinate.
- Who are going to run which of the 8 cosmological halos for each code group, starting with DM-only runs? (Also check out [this prototype](#) where we intend to collect such information.)

Code	1e10q	1e10v	1e11q	1e11v	1e12q	1e12v	1e13q	1e13v
ART-I			DC					
ART-II			SL		SL			
ENZO	CS	CS	JK					
GADGET-2/3 (various versions)	JO/MR	JO/MR	MR/AH/KT	MR	MR	MR		
GASOLINE			SS				SS	SS
PKDGRAV-2			JG/MK					
RAMSES			RT					

[4. Other News: Flagship Paper, Mini-Workshop Opportunity, etc.](#)

(1) Flagship Paper (Paper "1"):

As of November 2013, the Flagship paper has been accepted by the ApJS, and is going through an editorial process (to be published in February 2014). The current revised version of the manuscript is available on the [arXiv](#), and will be accordingly updated with the published version soon. Most notably in this revision, the science-oriented Working Group descriptions have been taken out from the Appendix and moved to the Project [website](#). The press releases by UC-HiPACC and UCSC are in the following locations. Joel Primack and Piero Madau thank Trudy Bell (UC-HiPACC) and Tim Stephens (UCSC) for their effort: [UCSC Press Release \(EurekAlert\)](#), [UC-HiPACC Press Release \(EurekAlert\)](#), [UC-HiPACC AstroShort](#)

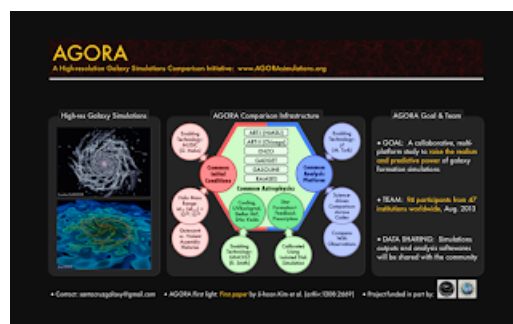
(2) Mini-workshop Opportunities for Individual Working Groups:

UC-HiPACC@UCSC and KIPAC@Stanford are willing to provide funding opportunities for working group leaders to organize mini-workshops (< 10 people) at either place as their science results come in, before the 3rd AGORA Workshop in 2014 (see below). Please feel free to talk to the project coordinator at [santacruzgalaxy\(at\)gmail.com](mailto:santacruzgalaxy(at)gmail.com) if you wish to organize one. A possible yt-AGORA mini-workshop is being planned in mid to late April.

(3) One-slide Introduction of AGORA:

Please click the link to see a larger image and more information. Feel free to use this slide in your talk.

sites.google.com/site/projectagoraworkspace/home/collaboration-blog/one-slidesummaryofagora



(4) First Article Using the AGORA Initial Conditions:

First article using the AGORA initial conditions is out: Hopkins et al. (2013, [arXiv](#)). Specifying the initial conditions utilized, we expect to naturally build a "library" of AGORA simulations. This practice will enable easier comparison with different codes in the future.

(5) Workshop in 2014 (very tentative):

Annual Santa Cruz Galaxy Workshop (Aug. 11–15, 2014) + 3rd AGORA Workshop in UCSC (Aug. 15–17, 2014)

(6) Useful Links:

- Collection of softwares relevant to AGORA effort: [here](#)
- Quick links to the important pages in AGORA Workspace: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)

